Fixed-Point Binary Numbers

1. What Are Fixed-Point Numbers?

A **fixed-point number** is a representation of real numbers using a fixed number of bits split into two parts:

- **Integer part**
- **Fractional part**

Unlike floating-point formats, the binary (radix) point stays in a fixed position. This makes fixed-point arithmetic simpler and faster in systems where hardware floating-point support is limited.

Applications

- Embedded systems
- Digital signal processing (DSP)
- Audio/video encoding
- Control systems

2. Format Structure

A fixed-point number typically looks like:

```
[sign bit (if signed)] [integer bits].[fractional bits]
```

Let:

- I = number of integer bits
- F = number of fractional bits
- $N = I + F$ = total number of bits (not including the binary point)

Example (Unsigned, 8-bit total = 4 integer + 4 fractional):

```
Binary:  0011.1010
```

- Integer part: $0011 = 3$
- Fractional part: $.1010 = 0.5 + 0.125 = 0.625$
- Total value: **3.625**

3. Binary Fraction Conversion

Each bit after the binary point represents a negative power of 2:

Bit Position	Weight	Example Contribution
1st	$2^{-1} = 0.5$	$1 \times 0.5 = 0.5$
2nd	$2^{-2} = 0.25$	$1 \times 0.25 = 0.25$
3rd	$2^{-3} = 0.125$	$0 \times 0.125 = 0$
4th	$2^{-4} = 0.0625$	$1 \times 0.0625 = 0.0625$

Example: $0.1101 = 0.5 + 0.25 + 0 + 0.0625 = \mathbf{0.8125}$

4. More Examples

Unsigned Format (4 Integer + 4 Fractional)

- Binary: 0100.1100
- Decimal: $4 + 0.5 + 0.25 = \mathbf{4.75}$

Signed Format Using Two's Complement (4 Integer + 4 Fractional)

- Binary: 1101.0110
- Integer (two's complement): $1101 = -3$
- Fraction: $.0110 = 0.25 + 0.125 = 0.375$
- Decimal: **-2.625**

5. Range and Precision

For 8-bit unsigned (4 integer + 4 fractional):

- **Range:** 0 to $(2^4 - 1) + (1 - 2^{-4}) = 15.9375$
- **Precision:** Smallest difference $= 2^{-4} = \mathbf{0.0625}$

For signed (Two's complement, 4 integer bits):

- **Range:** -8.0 to $+7.9375$

6. Pros and Cons

Advantages	**Disadvantages**
Simple and fast in hardware	Fixed precision (no automatic scaling)
Low power consumption	Manual handling of overflow/underflow
Deterministic behavior	Less flexible than floating-point

7. Summary

- Fixed-point numbers use a static binary point position.
- Useful in low-resource environments where speed, power, and predictability matter.
- Fractional bits use negative powers of 2.
- Two's complement handles negative numbers.
- Careful design needed to balance range and precision.